

TREATMENT AND EMERGENCY RECOMMENDATIONS FOR Very Long Chain Acyl-CoA Dehydrogenase Deficiency (VLCAD Deficiency)

PATIENT: _____ MRN: _____ DOB: _____

This patient has VLCAD Deficiency and is at risk for serious complications from this condition.

**Please see last section of this letter for
Emergency Treatment during Decompensation**

Definition: VLCAD Deficiency is an inherited problem of fatty acid metabolism. Persons with this condition appear totally normal but can have problems when stressed by infection, fasting, or other physiological challenges.

Signs of Decompensation: Vomiting, lethargy, decreased state of consciousness, low blood sugar, elevated ammonia, elevated CPK, myoglobinuria, rapid heart rate, shortness of breath, and coma.

Treatments:

- A. ***Prevention of decompensation:***
1. Avoid fasting for long periods ("long periods" are relative and depend upon age).
 2. Use frequent feedings containing carbohydrate, low in fat, and supplemented with Medium Chain Triglycerides (MCT).
 3. Give L-carnitine to replace body carnitine that is lost through the urine with the excretion of fatty acids (L-carnitine dose of 100 mg/kg/24 hours may be taken in 2, 3, or 4 doses throughout the day. Carnitor® [TRADE NAME] comes in a liquid form (100 mg/ml, tablets (330 mg), and capsules (250 mg).
- B. ***Treatments during an illness*** (colds, flu, ear infections, etc):
1. Infant formulas such as Pregestimil and Lipistart are designed to be low in fat and high in MCT.
 2. Encourage fluids containing carbohydrate:
 - i. Infant formula

- ii. Fruit juices
 - iii. Gatorade or ginger ale
 - 3. Observe urine output for hydration status and color for signs of rhabdomyolysis.
 - 4. IF YOUR CHILD IS LETHARGIC (SLEEPY AND POORLY RESPONSIVE) TAKE HIM/HER TO THE EMERGENCY ROOM FOR IV GLUCOSE.
 - 5. During illnesses you may give one or two extra doses of L-carnitine per day (remember that this may cause loose stools and diarrhea, so if diarrhea is part of the illness this step can be left out).
- C. ***Treatment during decompensation*** (Lethargy, vomiting, low blood sugar, elevated CPK, decreased state of consciousness, and/or coma. Please note that not all of these symptoms need to be present):
- 1. IV fluids including dextrose to keep blood glucose elevated.
 - 2. Glucose (D10 or greater in a $\frac{1}{4}$ to $\frac{1}{2}$ normal saline and may add potassium) to keep the blood glucose 110 to 170 mg/dL.
 - i. Use D10 even if the blood sugar is normal, and for maintenance.
 - ii. Use D25, 2 ml/kg for hypoglycemia.
 - 3. For metabolic acidosis use sodium bicarbonate, 1 mEq/kg, IV push.
 - 4. IV carnitine (200 mg/ml), if unable to take by mouth.
 - i. Loading dose is 100-200 mg/kg of body weight given as a bolus over 60-120 minutes,
 - ii. To be followed by 100-200 mg/kg per 24 hours in 6 divided doses or by continuous drip.
 - 5. Laboratory studies: Ammonia, CK, liver enzymes and blood gases.
 - 6. Treat the underlying and precipitating illness (infection, etc), and obtain appropriate additional laboratory studies to establish the cause of the physiological challenge.
 - 7. **Notify the metabolic specialist immediately (after business hours this number allows you to speak with the geneticist on call:**
Kentucky Childrens Hospital
Division of Genetics and Metabolism
R. Stephen Amato, MD, PhD
859-323-0396
Kentucky Clinic – J420
740 S. Limestone Street
Lexington, KY. 40536-0284
(After Hours: Please call 1-800-888-5533)